

General Entomology

ZOO 1203

Lesson 08

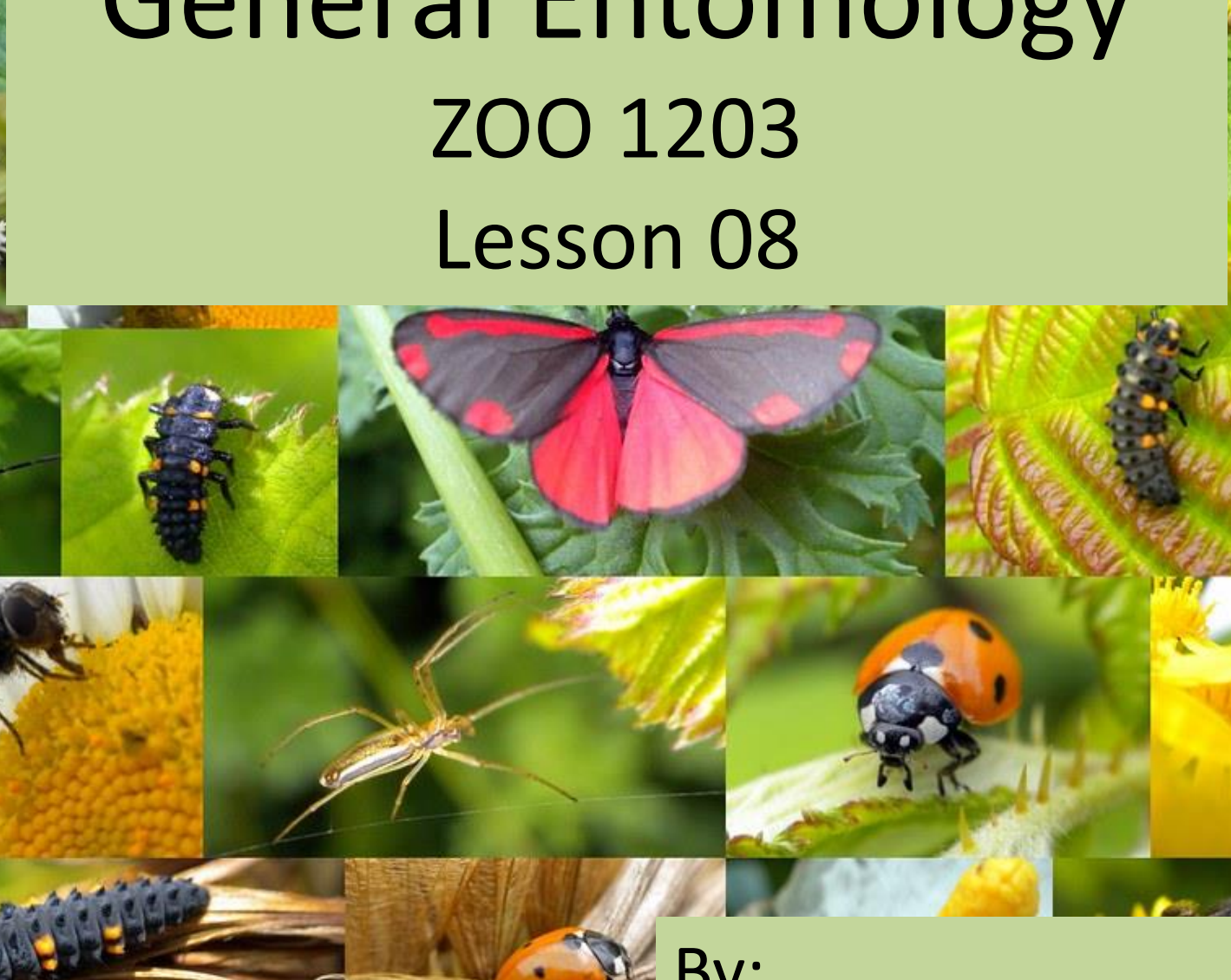


By;
Dr. Dilani Hettiarachchi

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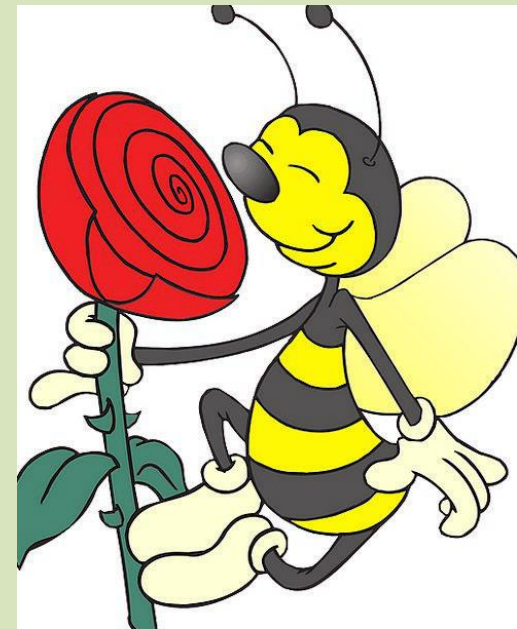
Lesson Objectives

After completion you should be able to;

- Name the available sensory receptors
- Explain each of the sensory receptors
- Describe the visual structures of insects
- Describe the host finding mechanism of insects

Sensory systems and behaviour

- Insects are successful due to being able to sense and interpret their surroundings and discriminate on a fine scale
- Insects can identify, differentiate and respond selectively to cues
- Many insects have well-developed vision
- Also use olfaction and hearing too



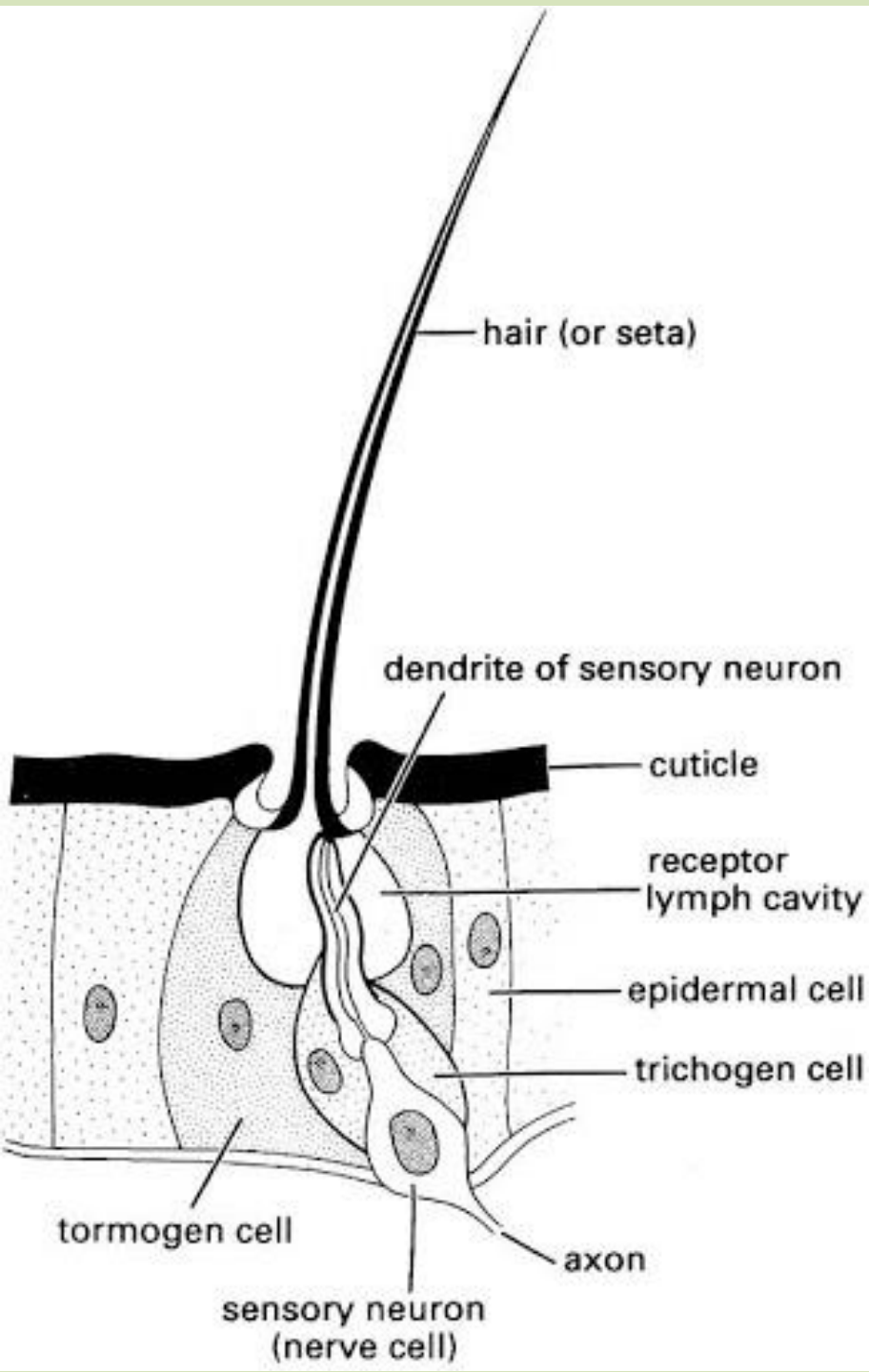
Types of sensory organs

1. Mechanoreceptors
2. Auditory receptors
3. Chemo-receptors
4. Thermo-receptors
5. Photo-receptors

Mechanoreceptors

- Cuticular projections
- Called microtrichia, hairs, bristles, setae or macrotrichia
- Sensilla → trichoid sensilla (hair-like sense organs)
- Developed from epidermal cells
- act as **tactile** receptors, detecting movement of objects in the environment, or they may provide **proprioceptive** cues (sensory input about the position or orientation of the body and its appendages)

- These receptors are innervated by one or more sensory neurons
- response to stretching, bending, compression, vibration, or other mechanical disturbance
- Mainly four types;
 1. Trichoform sensilla
 2. Campaniform sensilla
 3. Stretch receptors
 4. Pressure receptors



Trichoform sensilla

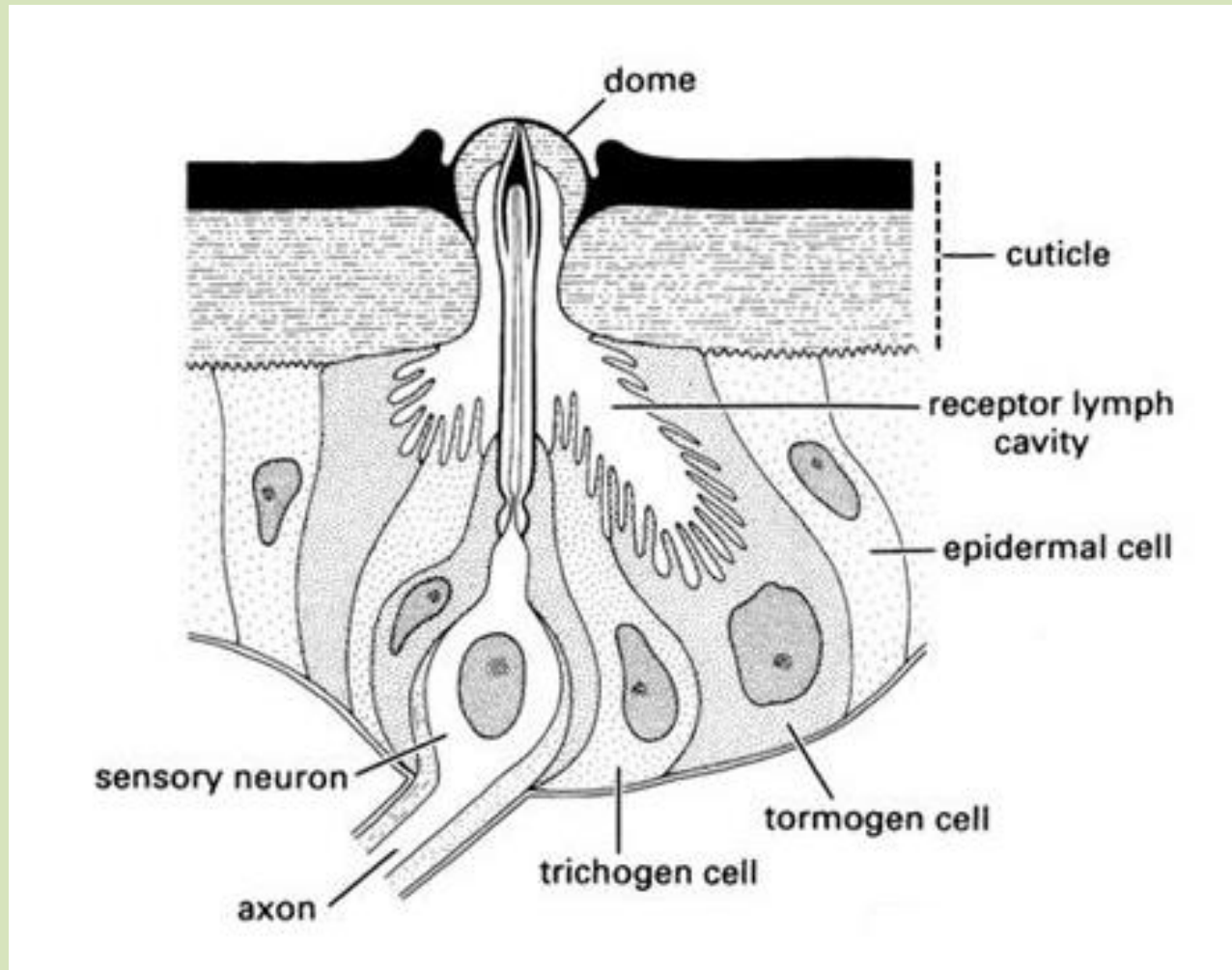
- the simplest mechanoreceptors
- **tactile hairs** (setae) that are innervated by a sensory neuron
- Three cells are involved in production;
 1. trichogen cell – makes the conical hair
 2. tormogen cell – makes the socket
 3. sensory neuron – connects to the CNS

- Dendrites of the neuron attach near the base of the hair and generate a nerve impulse whenever they detect movement
- often found behind the head, on the legs, or near joints where they respond to movements of the body.

Campaniform sensilla

- flattened oval discs
- They respond whenever mechanical stress causes the exoskeleton to bend

- found throughout the body — especially on the legs, near the base of the wings, and along sutures where two sclerites of the exoskeleton meet



Stretch receptors

- multi-polar neurons that usually accompany muscle or connective tissue
- commonly embedded in inter-segmental membranes and in the muscular walls of the digestive system
- Eg. oviposition may begin when mature eggs stimulate stretch receptors in the reproductive system.

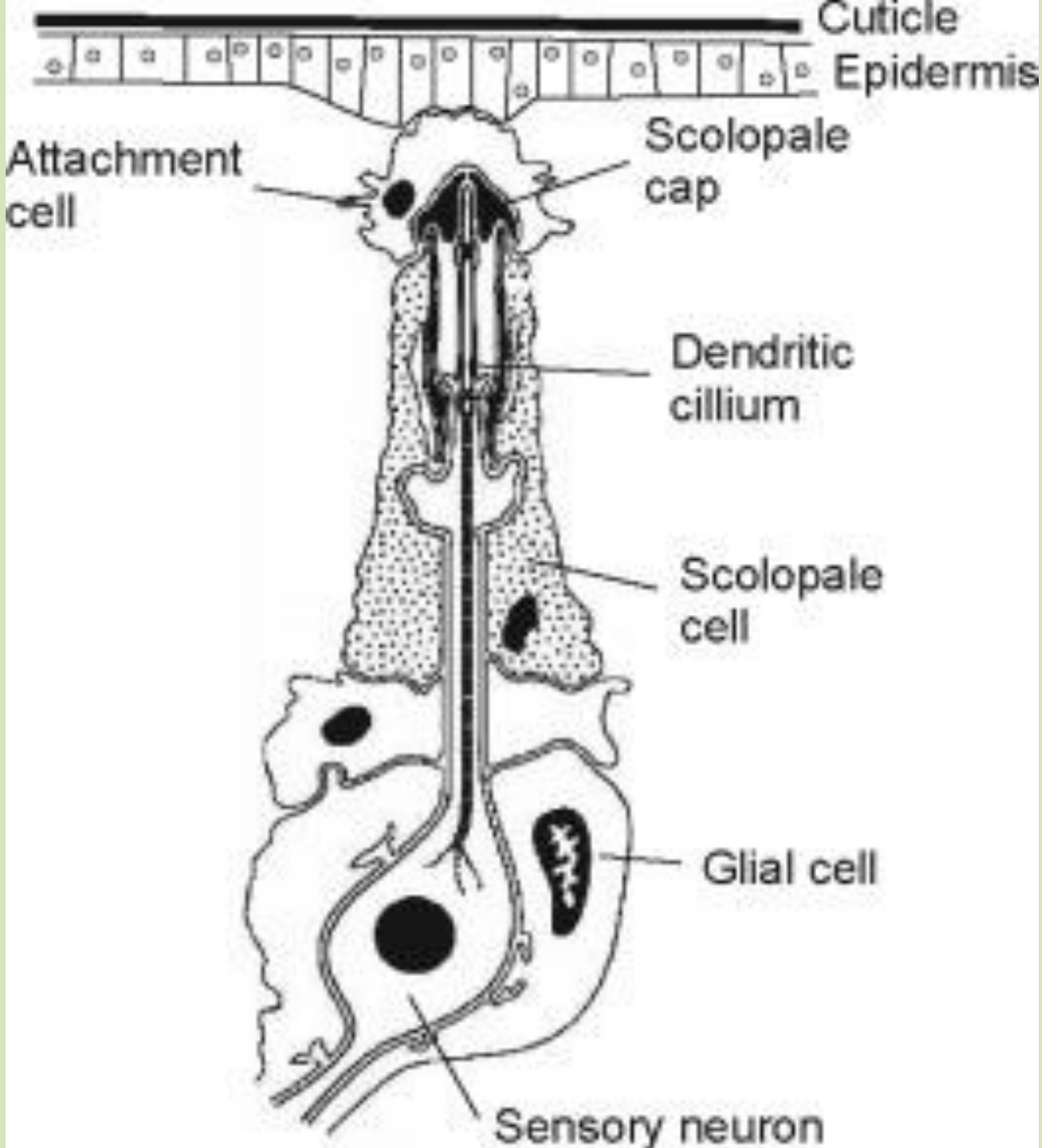
Pressure receptors

- provide sensory information about an aquatic insect's depth in the water

- usually associated with a cushion of air against the body or within the tracheal system

- In addition to touch insects also respond to hardness of the food
- Sensilla in the heavily sclerotized cusps of the mandibles &/or in maxillae
- Touch receptors are widely distributed over all parts of the body
- Individual identity of the receptors must remain by the time it reaches CNS

- These mechanoreceptors together form chordotonal organs
- **Chordotonal organs** →
 - include several types of mechanoreceptors
 - one or more bipolar neurons bridge a gap between two internal surfaces of the exoskeleton
 - Each neuron is usually accompanied by two other cells which form a sheath (the scolopale cell) and a point of attachment (the cap cell).
 - Together, these three cells create a unit called a **scolopidia**



- Each cell is composed of

1. Neurone
2. Scolopale cell
3. Cap cell

- These scolopidia may occur singly or in groups
- Common types of chordotonal organs include:
 - **Subgenual organs** — located in the legs of many insects. very sensitive to substrate vibrations
 - **Tympanal organs** — lie beneath a drum-like membrane (the tympanum) where they respond to sound vibrations
 - **Johnston's organs** — found within the pedicel of each antenna. In some insects, they function as a proprioceptors, supplying information on position or orientation of the antennae

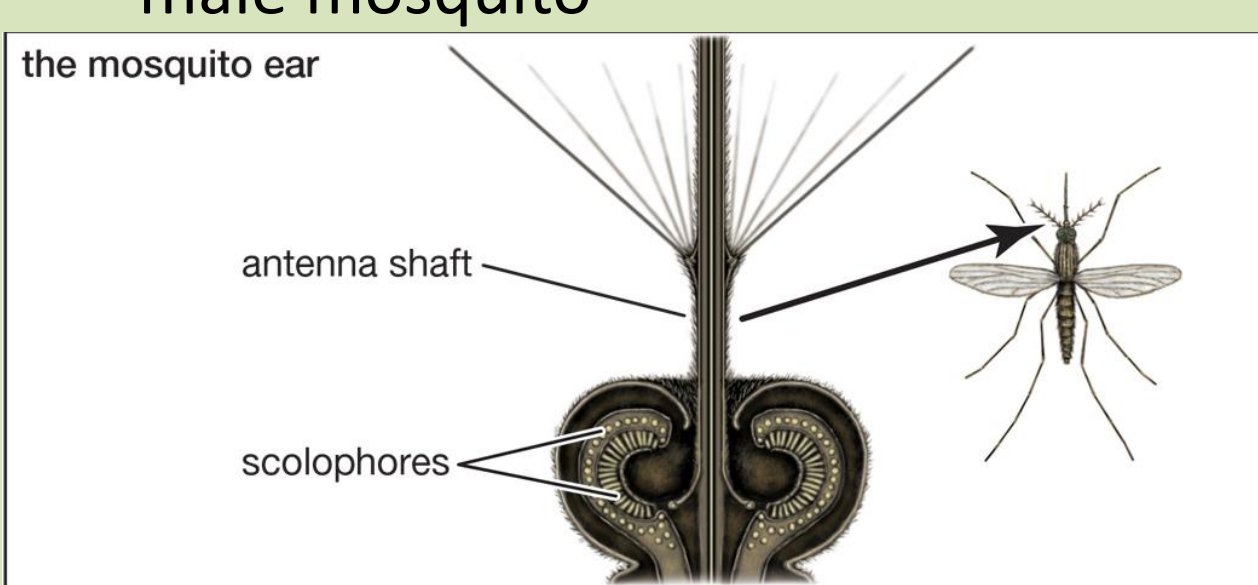
Auditory receptors

- Sound is a pressure fluctuation transmitted in a wave form via movement of the air or the substrate, including water
- Some insects can detect as low as 1 – 2 Hz to ultrasound frequencies such as 100 kHz
- They have intraspecific acoustic communication
eg: male grasshopper produce species specific sound (“songs”) that non-singing female detect
- Hearing also avoid predation
eg: insectivores bats which use ultrasound in hunting

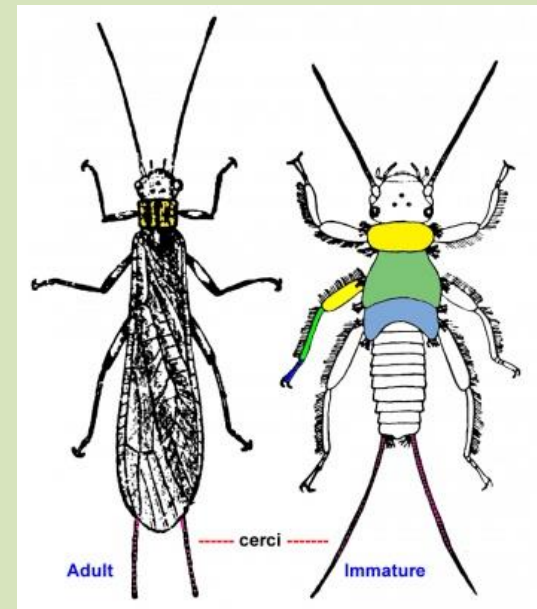
- katydids, crickets, and cicadas produce distinct sounds
- Hearing organs have been found in insects at various locations on the thorax and abdomen and, in one group (mosquitoes), on the head.
- hearing is known to exist in only a few orders:
 - Orthoptera (crickets, grasshoppers, katydids),
 - Homoptera (cicadas),
 - Heteroptera (bugs, waterboatman)
 - Lepidoptera (butterflies and moths) and
 - Diptera (mosquitoes)

Hair sensilla

- These are **Trichoform sensilla**
- primarily mediate the sense of touch and that their response to sound waves is only incidental to that function
- Delicate tactile hairs are present in plumose antenna of male mosquito



- Elongated trichoid sensilla respond to vibration produced by near-field sounds
- Eg: caterpillars of the noctuid moth have hairs that are 0.5 mm long but can detect vibrations of 150 Hz
- Cerci of many insects, especially crickets, are sensitive to air current → approach of predators, parasitic insects or potential mate

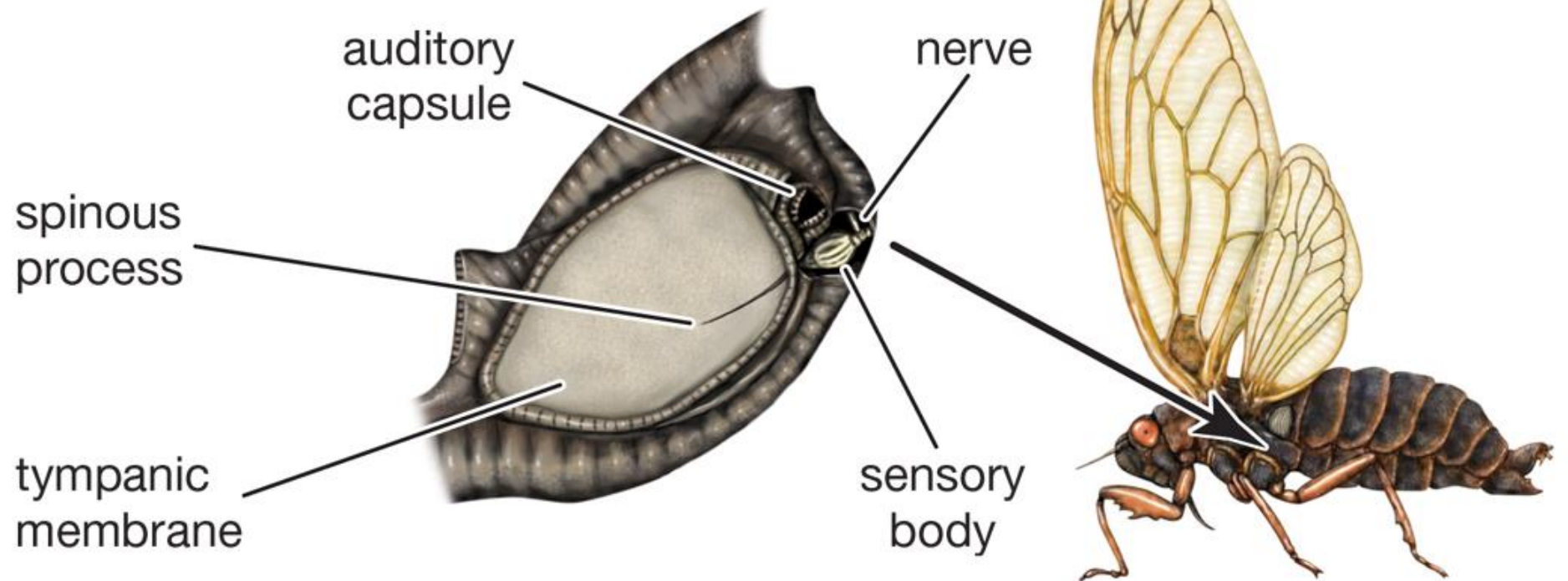


- Specialized cells that receive vibrations are subcuticular mechanoreceptors called **chordotonal organs**
- All adult insects and many larvae have Johnston's organ lying within the pedicel, the second antennal organ
- Detection of vibrations involve subgenual organ located in the proximal tibia of each leg
- Subgenual organs are found in most insects except Coleoptera and Diptera

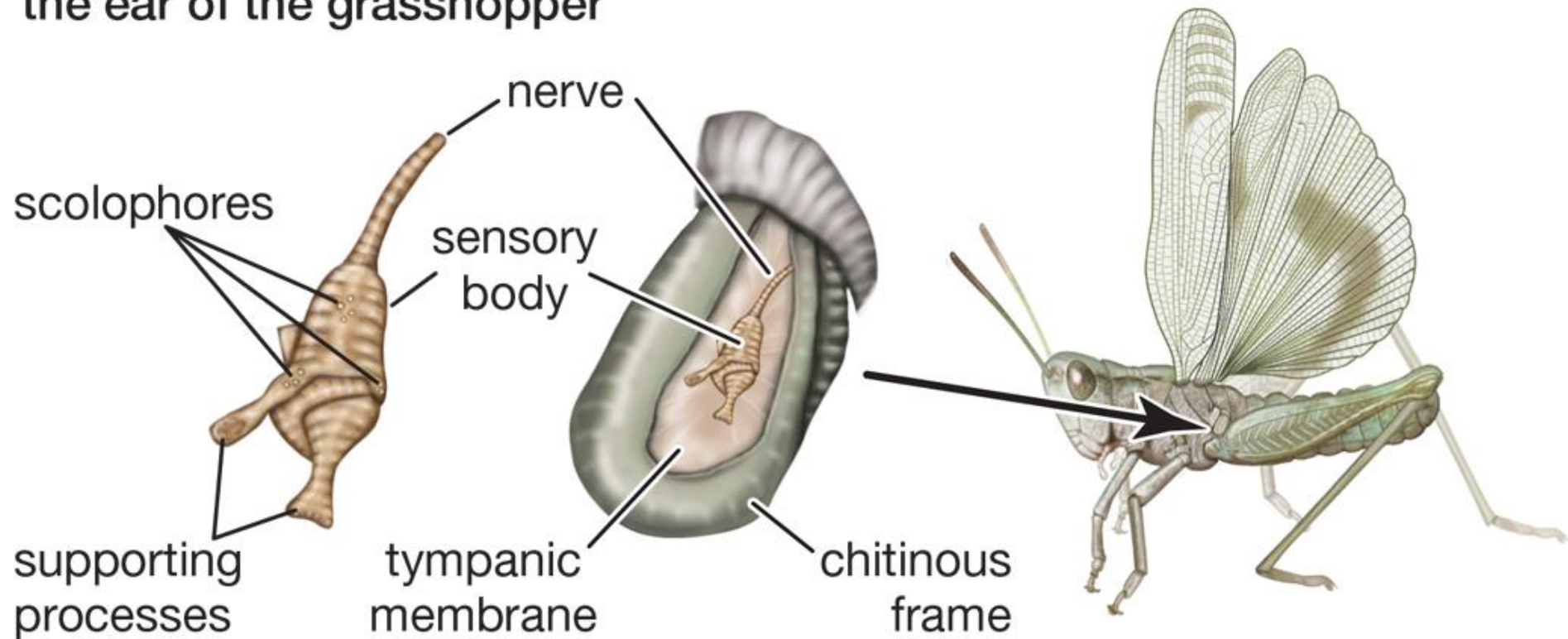
Tympanal organs (Tympani)

- Contains scolophore organs/ Chordotonal organs

the ear of the cicada *Magicicada septendecim*



the ear of the grasshopper



- Tympanum organs are located in;
 - prothoracic leg of orthopterans
 - metathoracic legs of mantids
 - abdomen of cicads
 - wings of certain moths and lacewings
 - etc.....

- Tympanum vibrations are detected by three chordotonal organs;
 - Subgenual organ → present in all legs
 - Intermediate organ } found only on the fore legs in
 - Crista acustica } conjunction with the tympanum
- Five orders have insects that have the ability to detect ultrasounds;
 - Neuroptera – Lacewings
 - Coleoptera – beetles
 - Mantodea – Praying mantids
 - Lepidoptera – moths
 - Orthoptera – locusts, katydids and crickets

Chemo-receptors

- Insects profoundly use chemicals in communication
- Chemical senses can be divided into;
 - taste/contact/gustatory – detection of aqueous chemicals
 - smell/ distant/olfactory – detection of air borne chemicals
- Trap chemical molecules which are transferred to a site for recognition
- Many contact/taste receptors occurs in the mouth parts and on the ovipositor

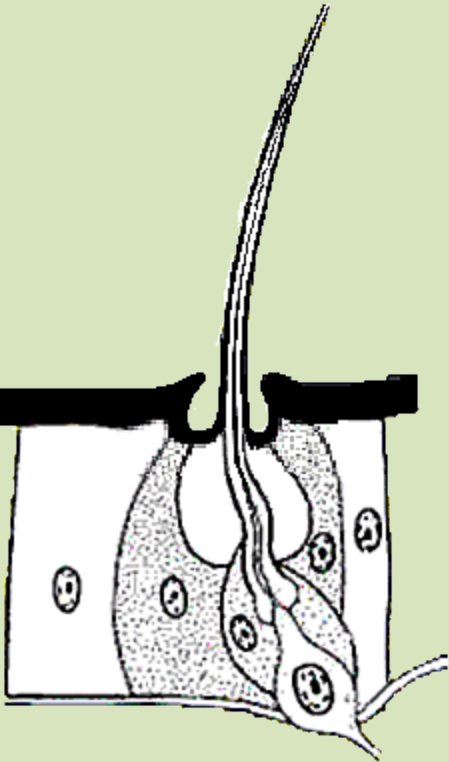
- Insect chemoreceptors are sensilla with one or more pores
- Two classes of sensilla; Uniporous and Multiporous
- Uniporous → one pore at the apex or central appearance from hair to pegs, plates or simply pores

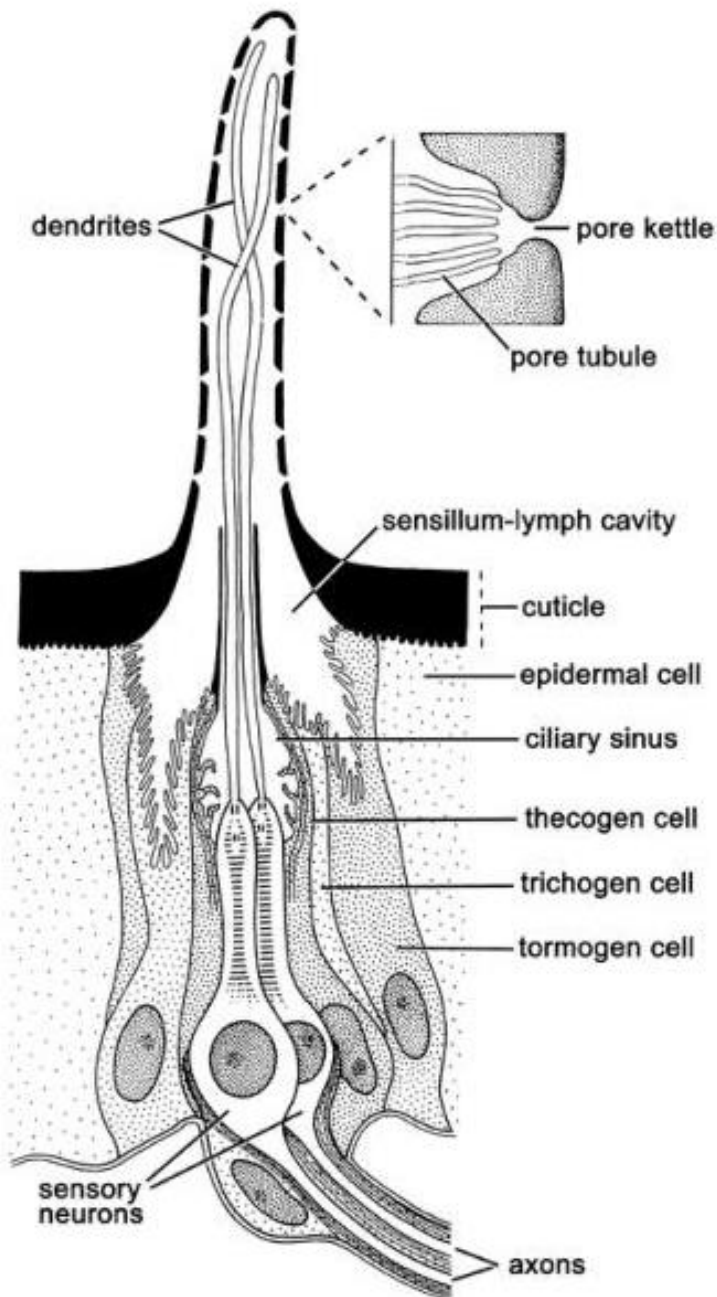
Relatively thick walls

Hair /peg contain a chamber

Detect chemicals by contact

There are evidence for some olfactory function





- Multiporous →
- hair or peg like setae
- many round pores or slits in the thin walls
- pores lead into a chamber known as pore kettle
- there are pore tubules which run inwards to meet with multi-branched dendrites

- Location of tastes are common in;
 - Mouthparts especially the palp
 - The antennae (Hymenopterans)
 - Tarsi (many Hymenoptera and Diptera)
 - General body surface
- Location of organs of smell is in;
 - The antennae
 - Mouthparts in some insects

Semiochemicals : Pheromones

- Many insect behavior rely on the sense of smell
- Chemical odours → semiochemicals
- Mainly for communication
- Used in both interspecific and intraspecific communication
- Intraspecific communication involves the use of pheromones
- First recognized in 1950's

- Pheromones;
“substances that are secreted to the outside by one individual and received by the second individual of the same species in which they release a specific reaction”
- Pheramones are volatile but sometimes are liquid contact chemicals
- Produced by exocrine glands derived from epidermal cells
- Scent organs may be located anywhere on the body

- Different type of pheromones that elicit different behaviours

1. Sex pheromones → sex attraction pheromones and courtship pheromones

2. Aggregation pheromones

3. Spacing pheromones

4. Alarm pheromones

5. Trail-marking pheromones

Semiochemicals : Kairomones, Allomones and Synomones

- Communication between different species is done by using allelochemicals
- Interspecific semiochemicals can be grouped according to the benefit they provide to producer and receiver

1. Kairomones

- ✓ benefit the receiver
- ✓ Disadvantage to the producer
- ✓ Eg- When pine is damaged by western pine beetle the tree produce Myrcene
But Myrcene is similar to aggregation pheromones and more beetles attract

2. Allomones

- ✓ benefit the producer
- ✓ Disadvantage to the receiver
- ✓ Eg-defensive or repellent chemicals

Certain orchids flower odors resemble the female sex pheromones of the wasp or bee species that acts as their specific pollinators

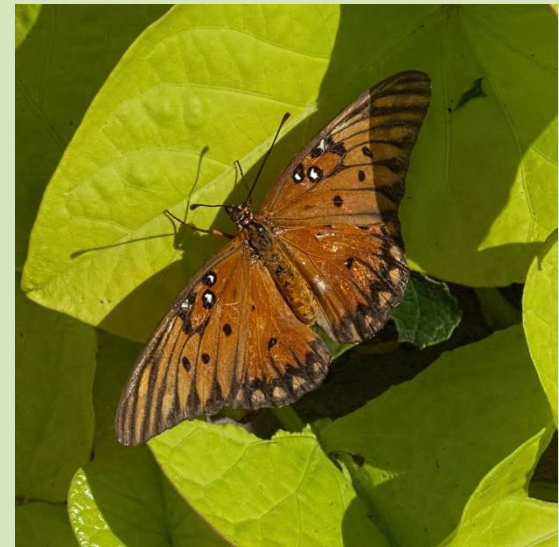
3. Synomones

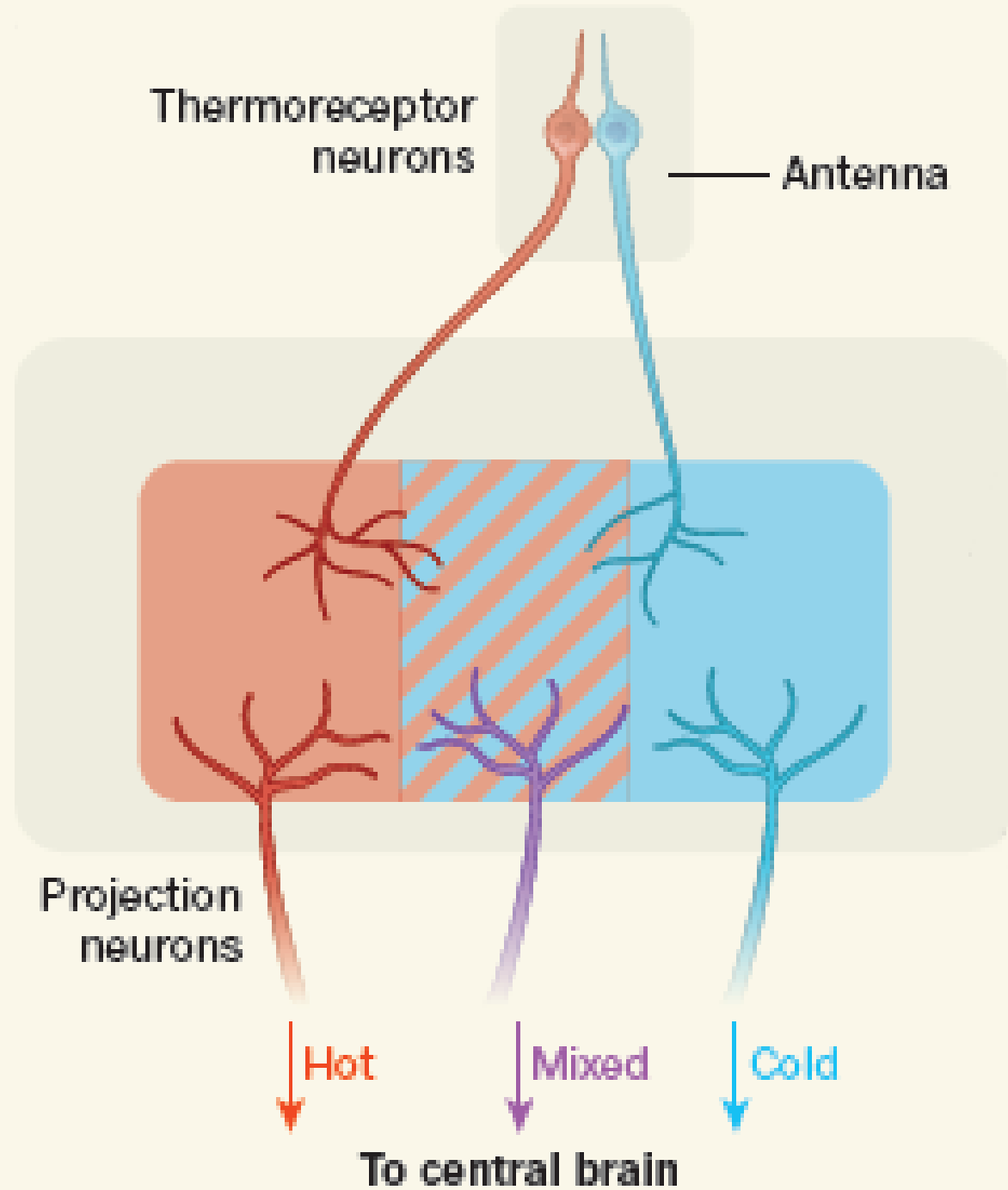
- ✓ Any substance produced by an individual of one species that benefits both the producer and the recipient which is of a different species.
- ✓ Eg – if a parasitoid is attracted to the pine trees due to Myrcene (kiremone) and attack the pine beetle

Thermo-receptors

- Process by which different level of heat energy in the environment and or in body is detected by organisms.
- Insects evolved mechanisms to survive.
- So, rapid detection of environment and body temperature is important for their survival
- Control of insect body temperature by physiological or behavioral means indicates presence of thermoreceptors
- In insects humidity and temperature receptors usually present together in a single sensillum

- These receptors are found in antennae of insects
- A typical thermo-hygro receptive sensillum contains about 2-5 sensory cells.
- Surrounding cells of sensillum deals with thermoreception.
- Eg – during early mornings butterflies can be seen spreading and directing their wings towards the sun





- Insects are able to detect the amount of water or vapour in the surrounding air
- They can search for preferred humidity or sources of liquid
- Humidity detectors are typically located on the antenna

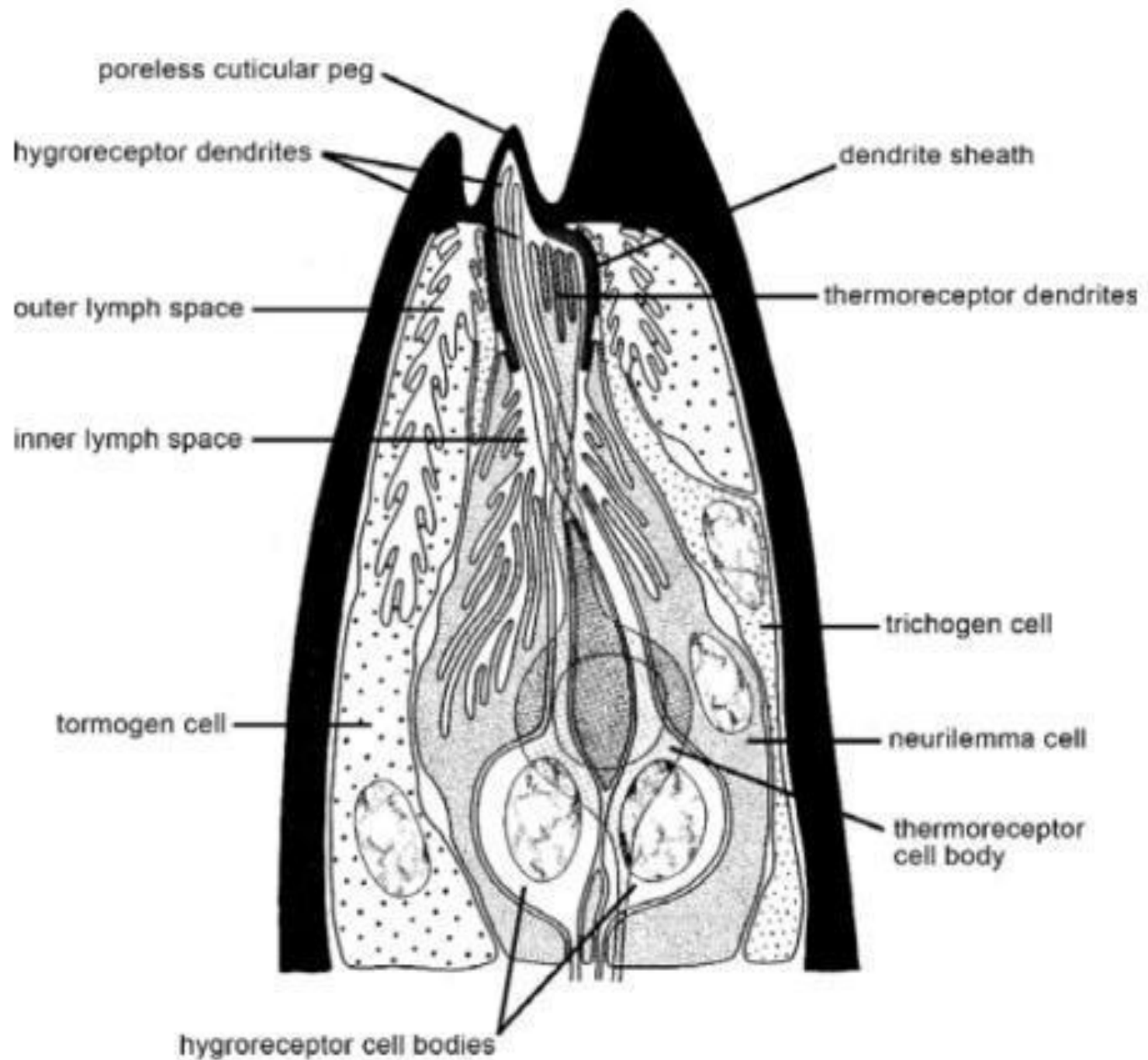


Photo-receptors

- Ability to perceive light in visible or near visible range of the electromagnetic spectrum
- Lenses need to focus the light onto photoreceptors
- Photoreceptors → cells containing light-sensitive molecules
- Three types of photoreceptive structures found in insects;
 1. compound eyes
 2. dorsal ocelli
 3. lateral ocelli (stemmata)

Stemmata (Lateral ocelli)

- Visual organ of holometabolous insects (complete metamorphosis)
- Located on the head
- Vary from single pigmented spot to 6 or 7 large stemmata each with numerous photoreceptors and associated nerve cells
- Simple stemma;
 - Simple light/dark sensors
 - Detect movements
 - Low acuity



clypeus

frons

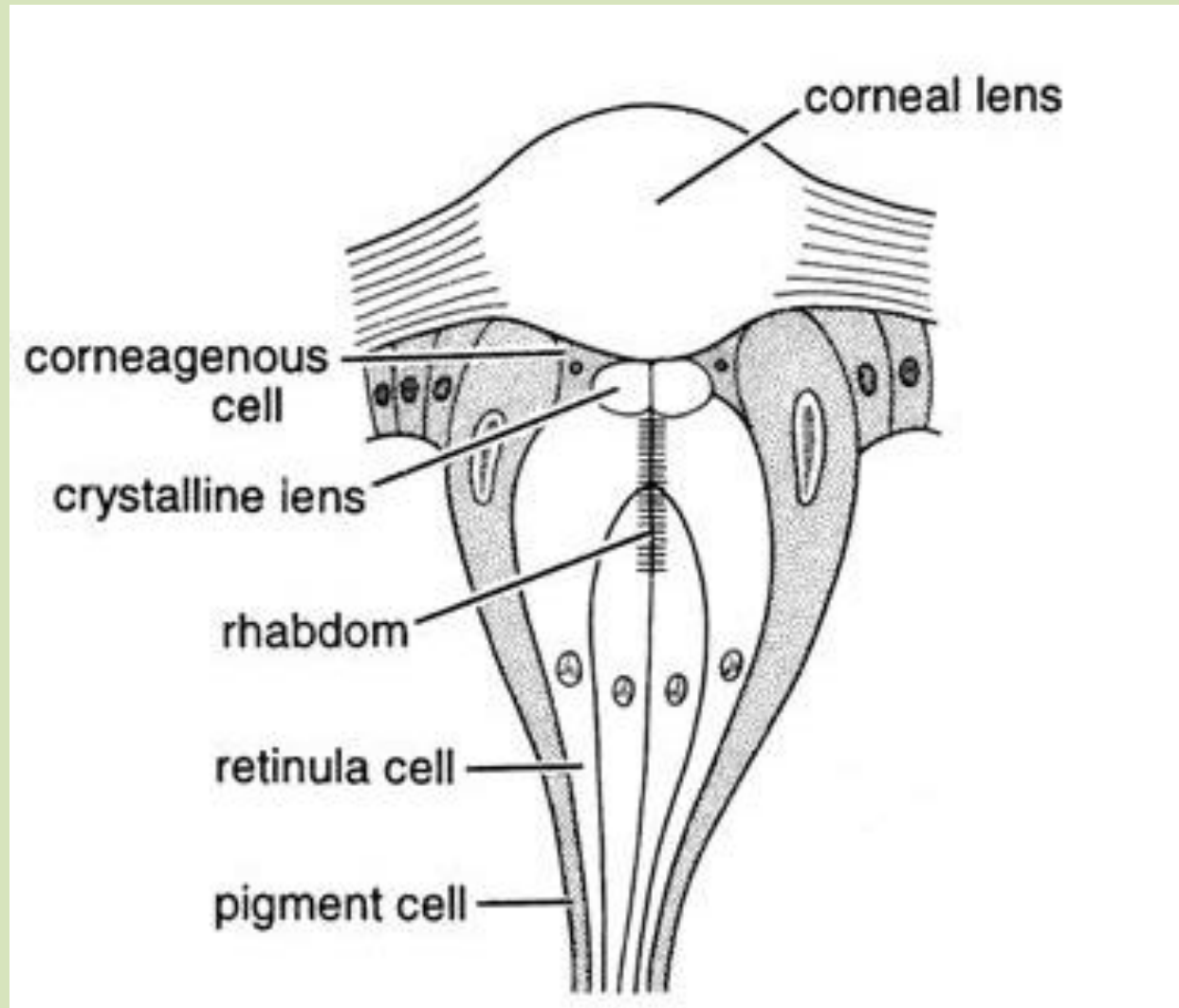
stemmata

labrum

mandibles

antennae

- Each stemma is pointed at different directions → insect see only few points
- Some caterpillars use scanning movements to see the whole picture



Ocelli (Dorsal Ocelli)

- commonly found in adults and in the immature stages (nymphs) of many hemimetabolous species.
- Dorsal ocelli appear as two or three small, convex swellings on the dorsal or facial regions of the head
- They differ from compound eyes in having only a single corneal lens covering an array of several dozen rhabdom-like sensory rods
- do not form an image or perceive objects in the environment

- they are sensitive to a wide range of wavelengths
react to the polarization of light and
respond quickly to changes in light intensity
- not independent visual organs
- never occur in species that lack compound eyes
- No exact function
- Adopted for the concentration of light and perception of
changes in intensity

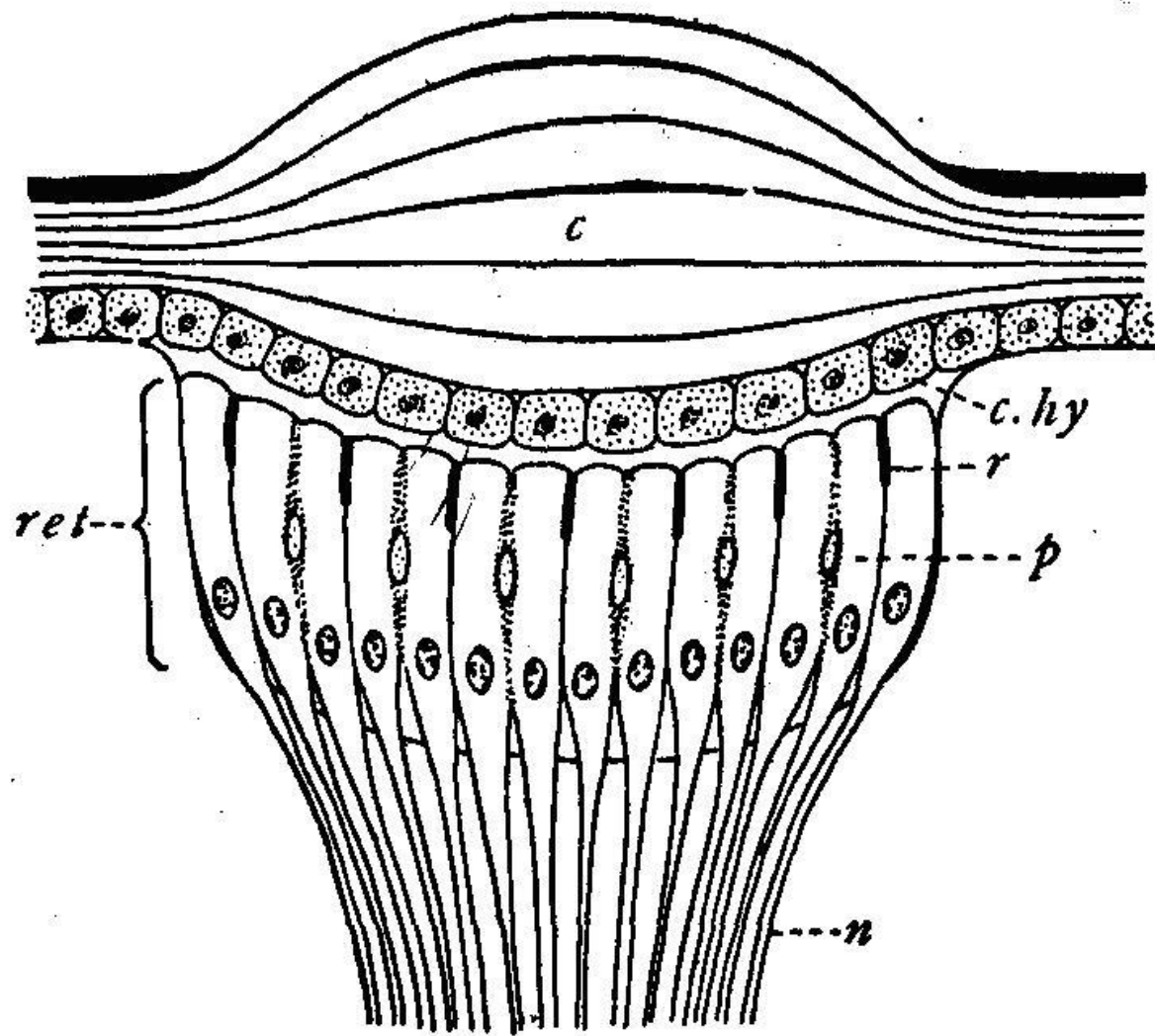


Fig. 154.—A diagram illustrating the structure of a primary ocellus; *c*, cornea; *c. hy*, corneal hypodermis; *ret*, retina; *n*, ocellar nerve; *p*, accessory pigment cell; *r*, rhabdom.



Compound eyes

- principle visual organs of most insects
- found in nearly all adults and in many immatures of ametabolous and hemimetabolous orders
- compound eyes are composed of many similar, closely-packed facets called **ommatidia**
- They are the structural and functional units of vision
- number of ommatidia varies considerably from species to species

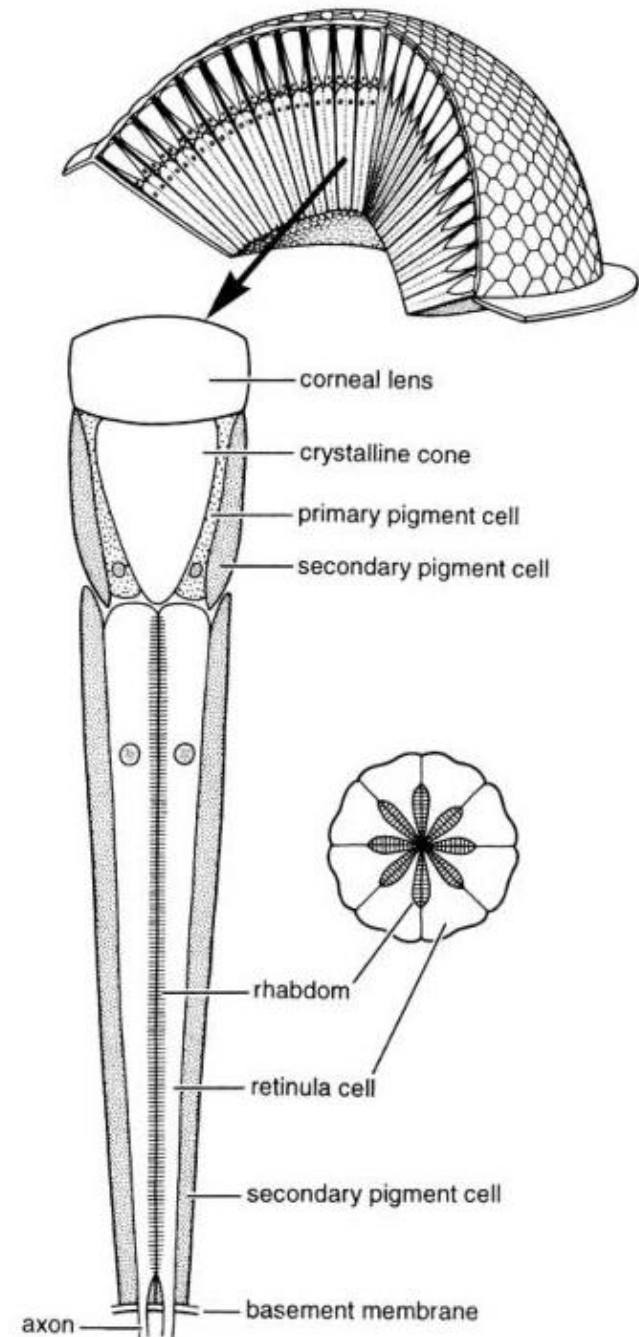


- Each ommatidium resembles a simple stemma

- Consisting of;
Optical; Light gathering part → corneal lens and crystalline cone

A sensory part; rhabdom, retinal cells (usually 8) and pigment cells

- Each cluster of retinal cells are surrounded by light-absorbing pigment cells



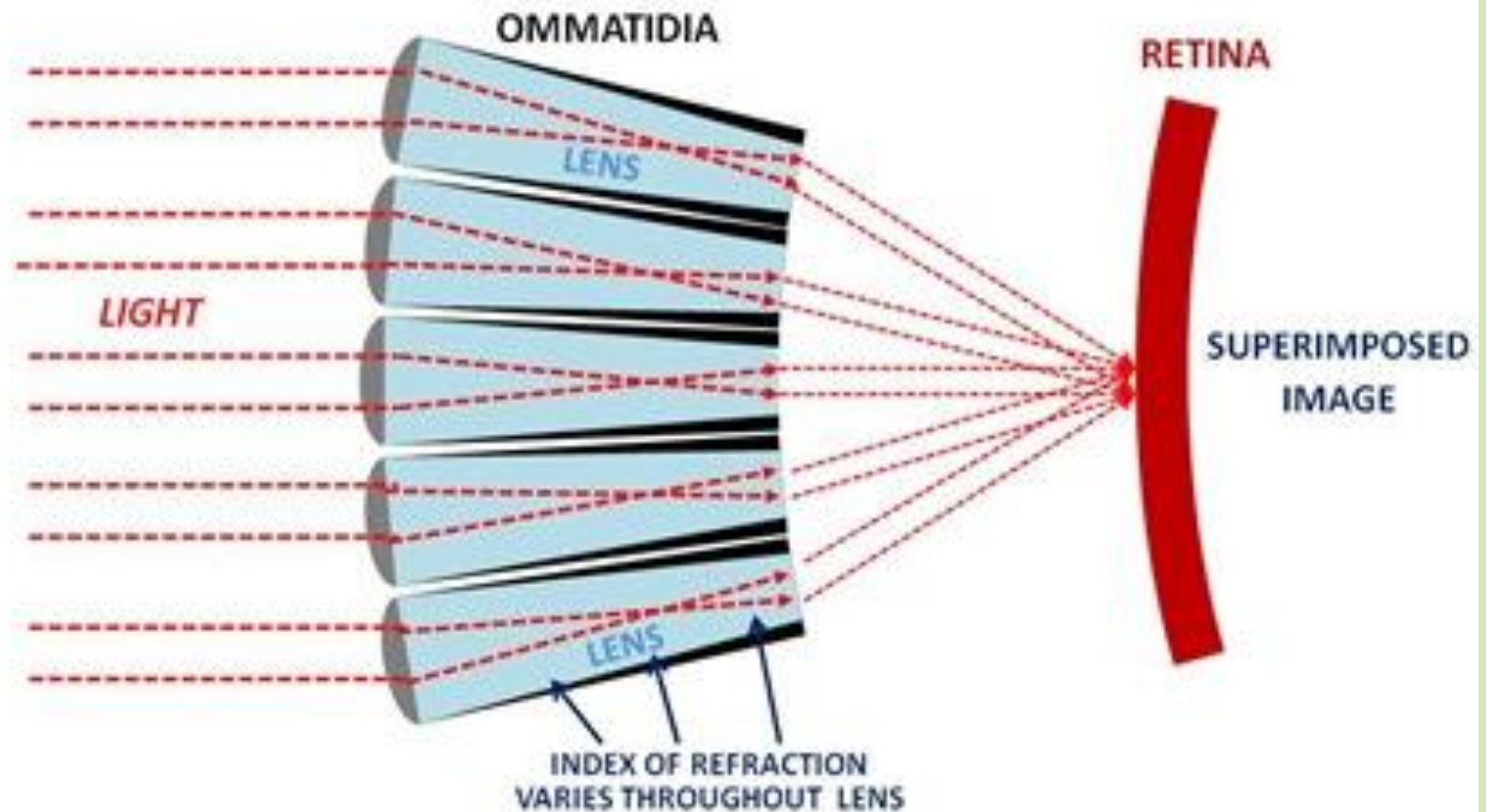
Apposition eyes

- Small lens diameter
- Omatidia are optically isolated from each other by the pigment cells

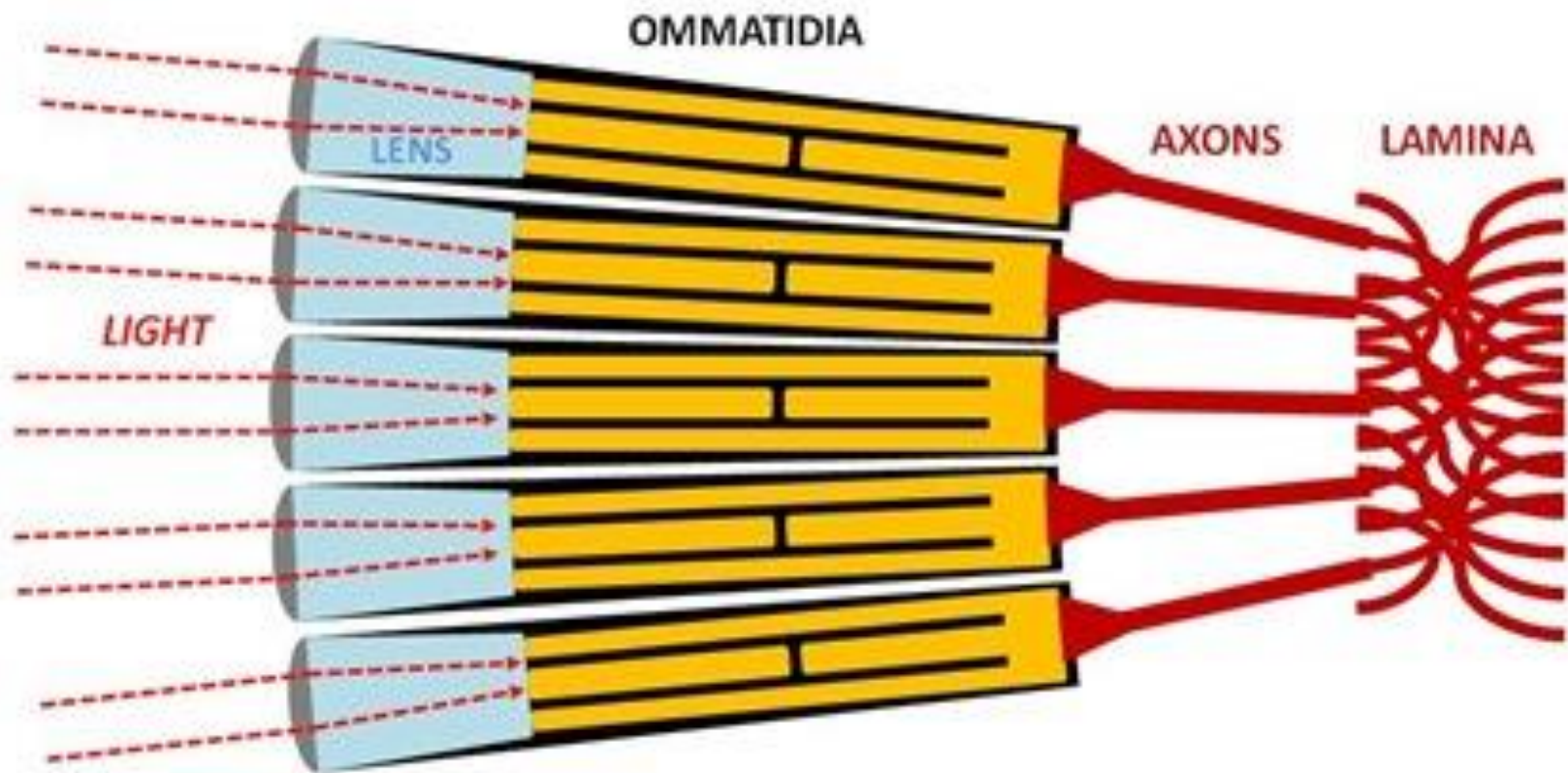
Superposition eyes

- Ommatidia are not optimally isolated from each other
- Corneal facet cells/lens are wide and clear
- Many lens co-operate to focus light on an individual rhabdom
- Light sensitivity is high

Refraction Superposition Eye



Apposition Compound Eye

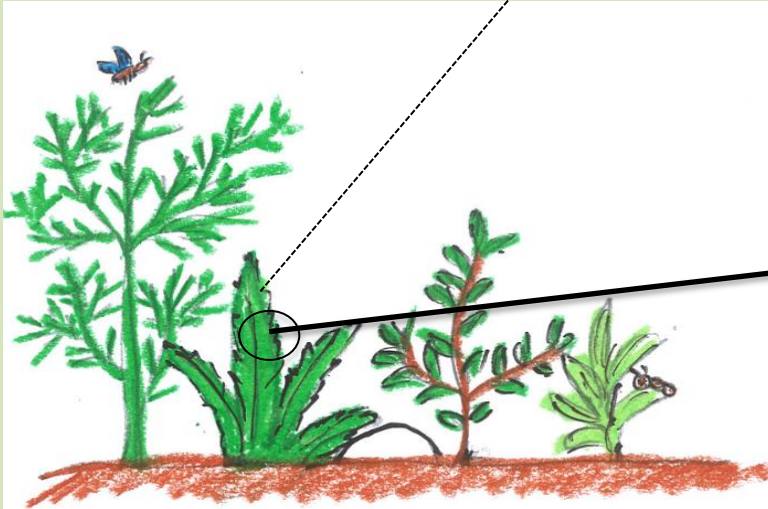


Involvement of receptors in food/host finding

Olfactory cues

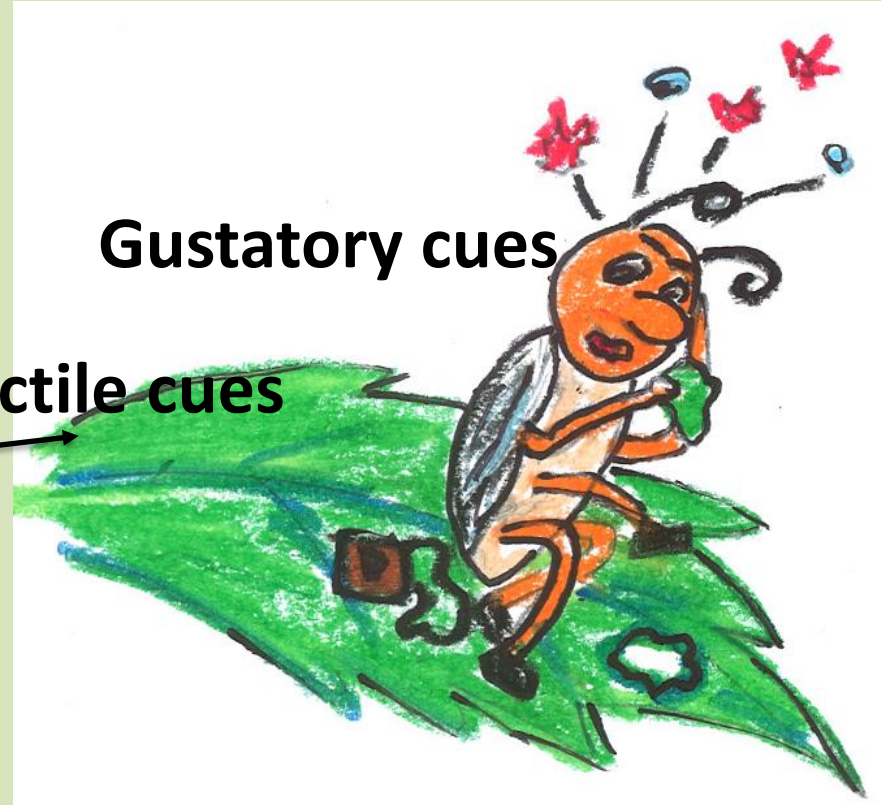


Visual cues



Gustatory cues

Tactile cues



Refer;

→ Light production by insects

→ Sound production by insects



THANK YOU!

